SCHEME THEORETIC DENSITY OF AN OPEN IMMERSION OF ALGEBRAIC SPACES CHARACTERIZED BY INJECTIVITY OF THE PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to the proof of the scheme theoretic image lemma for the closure computation; that lemma is in the same chapter and the argument for the converse direction is written out in full with the affine etale neighbourhood and the element f.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The characterization of scheme theoretic closure by the kernel of the structure map. Lemma lemma-scheme-theoretic-image, used at proof lines 3538-3540. Turns injectivity of the restriction of O_X -> j_*O_U into the closure statement.
- Etale localization of the injectivity of O_X -> j_*O_U. used at proof lines 3535-3538. Allows the statement to be tested on an arbitrary algebraic space V etale over X.
- Affine etale neighbourhoods detect non-injectivity of a map of sheaves. used at proof lines 3543-3547. Produces the nonzero f in A vanishing on the preimage, giving the contradiction.

Source reference. Original lemma 082X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10267–10267. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $j : U \to X$ be an open immersion of algebraic spaces over S. Then U is scheme theoretically dense in X if and only if $\mathcal{O}_X \to j_*\mathcal{O}_U$ is injective.*

Proof. If $\mathcal{O}_X \to j_*\mathcal{O}_U$ is injective, then the same is true when restricted to any algebraic space V étale over X. Hence the scheme theoretic closure of $U \times_X V$ in V is equal to V, see proof of Lemma 082X. Conversely, assume the scheme theoretic closure of $U \times_X V$ is equal to V for all V étale over X. Suppose that $\mathcal{O}_X \to j_*\mathcal{O}_U$ is not injective. Then we can find an affine, say $V = \mathrm{Spec}(A)$, étale over X and a nonzero element $f \in A$ such that f maps to zero in $\Gamma(V \times_X U, \mathcal{O})$. In this case

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the scheme theoretic closure of $V \times_X U$ in V is clearly contained in $\mathrm{Spec}(A/(f))$ a contradiction. $\square$